

Return vs. violation rate (final checkpoint, 312 episodes/seed; hollow = soft cost > 25/episode)



- Legend:
- SafeOT-typed (ours)
 - FDPI (constraint-based)
 - SAC-Lagrangian (penalty-based)
 - SAC + fixed penalty (penalty-based)
 - typed Pareto frontier (soft ≤ 25)
 - SDAC (multi-constraint SOTA)
 - SRCPPO (multi-constraint SOTA)
 - SAC-Pen coefficient sweep
 - FDPI p_f sweep
 - SAC-Lag cost-limit sweep